

MH3100 Real Analysis I

Midterm Practice 3 (Solutions)

AY 2026/2027, Semester 1 – QRS original

Quantitative Research Society @NTU

October 1, 2026

TIME ALLOWED: 60 minutes

Instructions to Students

1. This test contains **FOUR (4)** questions.
2. Answer **all** questions. The marks for each question are indicated below.
3. Write your answers in the space provided. If you need more space, continue on the facing page and clearly indicate that you have done so.
4. This is a **CLOSED BOOK** test.
5. You may use calculators. You may cite any results proved in the lectures and tutorials, **unless you are asked by the question to prove the result.**

Question	1	2	3	4	Total
Total Marks	6	6	6	6	24
Awarded Marks					

Question 1**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) If A and B are nonempty subsets of $\mathbb{R}$ with $A \subseteq B$ and B bounded above, then $\sup A \leq \sup B$.
- (b) (2 marks) The set $A = \{q \in \mathbb{Q} : q > 0 \text{ and } q^2 < 2\}$ has a supremum in $\mathbb{Q}$.
- (c) (2 marks) Every bounded monotone sequence of rational numbers converges to a rational number.

Solution

- (a) True

Proof. $B \neq \emptyset$ is bounded above $\therefore \sup B$ exists (AoC). For all $a \in A$: $a \in B \therefore a \leq \sup B \therefore \sup B$ is an upper bound of A . $A \neq \emptyset \therefore \sup A$ exists (AoC) and $\sup A \leq \sup B$ (least upper bound). $\square$

- (b) False

Proof. Suppose $s = \sup A \in \mathbb{Q}$. $1 \in A \therefore s \geq 1 > 0$; $s^2 \neq 2$ (no rational squares to 2).

Case $s^2 < 2$. Choose $n \in \mathbb{N}$ with $h := \frac{1}{n} < \min\{1, \frac{2-s^2}{2s+1}\}$ (Archimedean). Then $h \in \mathbb{Q}$ and

$$(s+h)^2 < s^2 + h(2s+1) < 2 \quad (h^2 < h < 1)$$

$\therefore s+h \in A$ and $s+h > s$, contradicting that s is an upper bound of A .

Case $s^2 > 2$. Choose $n \in \mathbb{N}$ with $h := \frac{1}{n} < \min\{s, \frac{s^2-2}{2s}\}$ (Archimedean). Then $s-h \in \mathbb{Q}$, $s-h > 0$ and

$$(s-h)^2 > s^2 - 2sh > 2$$

For all $q \in A$: $q^2 < 2 < (s-h)^2$ with $q, s-h > 0 \therefore q < s-h$. $\therefore s-h < s$ is an upper bound of A , contradicting that s is the least upper bound.

$\therefore s^2 < 2$, $s^2 > 2$ and $s^2 = 2$ are all impossible, so A has no supremum in $\mathbb{Q}$. $\square$

- (c) False

Proof. For $n \in \mathbb{N}$ let $S_n := \{k \in \mathbb{N}_0 : k \leq \sqrt{2} \cdot 10^n\}$; $0 \in S_n$ and $k < 2 \cdot 10^n$ for all $k \in S_n$. $\therefore S_n$ is finite and nonempty; let $k_n := \max S_n$ and $a_n := k_n/10^n \in \mathbb{Q}$. $k_n + 1 \notin S_n$ ($k_n = \max S_n$) $\therefore k_n \leq \sqrt{2} \cdot 10^n < k_n + 1 \therefore$

$$0 \leq \sqrt{2} - a_n < 10^{-n} \quad \text{for all } n \in \mathbb{N} \quad (*)$$

$10k_n \leq \sqrt{2} \cdot 10^{n+1} \therefore 10k_n \in S_{n+1} \therefore 10k_n \leq k_{n+1} \therefore a_n \leq a_{n+1}$. $(*) \implies a_n \leq \sqrt{2} < 2$ for all $n \therefore \{a_n\}$ is bounded and increasing, with terms in $\mathbb{Q}$.

Let $\varepsilon > 0$; choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$ (Archimedean). For all $n \geq N$:

$$|a_n - \sqrt{2}| < 10^{-n} \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon \quad ((*), 10^n \geq n)$$

$\therefore \sqrt{2} = \lim a_n$ (def. of limit), and $\sqrt{2} \notin \mathbb{Q}$ (no rational squares to 2). $\square$

Marking scheme (6 marks)**QRS suggested marking scheme.**

- **(2 marks)** Part (a): correct answer True.
- **(2 marks)** Part (b): correct answer False.
- **(2 marks)** Part (c): correct answer False.

No justification is required for full marks on this question; the proofs above are provided for the solutions build only.

Question 2**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks) Every infinite and bounded subset of $\mathbb{R}$ has a limit point.
- (b) (2 marks) The closure of a bounded subset of $\mathbb{R}$ is (sequentially) compact.
- (c) (2 marks) Every set with no limit points is (sequentially) compact.

Solution

- (a) True

Proof. Let $S \subseteq \mathbb{R}$ be infinite and bounded. For each n , $S \setminus \{a_1, \dots, a_{n-1}\}$ is infinite, hence nonempty. $\therefore$ distinct $a_1, a_2, \dots \in S$ can be chosen recursively. $\{a_n\}$ is bounded. $\therefore$ some subsequence has $a = \lim a_{n_k}$ (Bolzano–Weierstrass). Let $\varepsilon > 0$. There is K with $|a_{n_k} - a| < \varepsilon$ for all $k \geq K$. The a_{n_k} are distinct, so $a_{n_k} = a$ for at most one k . $\therefore$ some $k \geq K$ has $a_{n_k} \neq a$ (infinitely many $k \geq K$, at most one with $a_{n_k} = a$), so $a_{n_k} \in S \setminus \{a\}$ and $|a_{n_k} - a| < \varepsilon \therefore a$ is a limit point of S . $\square$

- (b) True

Proof. A bounded $\therefore$ there is $M > 0$ with $-M \leq a \leq M$ for all $a \in A$. Let $x \in \overline{A}$. There is $\{a_n\}$ in A with $x = \lim a_n$ (Sequential Criterion for Closures). $-M \leq a_n \leq M$ for all $n \therefore -M \leq x \leq M$ (OLT) $\therefore \overline{A}$ is bounded. Limit points of $\overline{A}$ are limit points of A (Limit Points of Closures), and these lie in $\overline{A} \therefore \overline{A}$ is closed. $\therefore \overline{A}$ is (sequentially) compact (Sequential Compactness of Closed and Bounded Sets). $\square$

- (c) False

Proof. Let $A = \mathbb{N}$. Suppose $a \in \mathbb{R}$ is a limit point of A . With $\varepsilon = \frac{1}{2}$: there is $x_1 \in A \setminus \{a\}$ with $|x_1 - a| < \frac{1}{2}$. With $\varepsilon = |x_1 - a| > 0$: there is $x_2 \in A \setminus \{a\}$ with $|x_2 - a| < |x_1 - a| \therefore x_2 \neq x_1$.

$$1 \leq |x_1 - x_2| \leq |x_1 - a| + |a - x_2| < 2|x_1 - a| < 1 \quad (\text{distinct } x_1, x_2 \in \mathbb{N})$$

This is a contradiction $\therefore A$ has no limit points.

Let $a_n := n$ and let $\{a_{n_k}\}$ be any subsequence. $n_k \geq k$ for all k (strictly increasing indices) $\therefore \{a_{n_k}\}$ is unbounded (Archimedean). $\therefore \{a_{n_k}\}$ diverges (convergent $\implies$ bounded) $\therefore A$ is not (sequentially) compact. $\square$

Marking scheme (6 marks)**QRS suggested marking scheme.**

- (2 marks) Part (a): correct answer True.
- (2 marks) Part (b): correct answer True.
- (2 marks) Part (c): correct answer False.

No justification is required for full marks on this question; the proofs above are provided for the solutions build only.

Question 3

(6 Marks)

In this question, A and B are sets of real numbers. **Write down the definition** of a limit point of a set, and **use the definition to show** that if x is a limit point of $A \cup B$, then x is a limit point of A or x is a limit point of B . *Do not use the Sequential Criterion for Limit Points.*

Solution

Definition (Limit Point). $a \in \mathbb{R}$ is a limit point of $S \subseteq \mathbb{R}$ if $\forall \varepsilon > 0 \exists x \in S \setminus \{a\} : |x - a| < \varepsilon$.

Method 1: Contrapositive

Proof. Suppose x is a limit point of neither A nor B ; we show it is not a limit point of $A \cup B$. Negated definition for A : there is $\varepsilon_1 > 0$ with $|y - x| \geq \varepsilon_1$ for all $y \in A \setminus \{x\}$. Negated definition for B : there is $\varepsilon_2 > 0$ with $|z - x| \geq \varepsilon_2$ for all $z \in B \setminus \{x\}$. Let $\varepsilon := \min\{\varepsilon_1, \varepsilon_2\} > 0$ and $w \in (A \cup B) \setminus \{x\}$. Then $w \in A \setminus \{x\}$ or $w \in B \setminus \{x\}$.

$$|w - x| \geq \varepsilon_1 \geq \varepsilon \quad \text{or} \quad |w - x| \geq \varepsilon_2 \geq \varepsilon \implies |w - x| \geq \varepsilon$$

$\therefore |w - x| \geq \varepsilon$ for all $w \in (A \cup B) \setminus \{x\}$. $\therefore x$ is not a limit point of $A \cup B$. $\therefore$ (contrapositive) if x is a limit point of $A \cup B$, then it is a limit point of A or of B . $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- **(2 marks)** Correct definition of a limit point, with the quantifiers in the correct order (“for every $\varepsilon > 0$ ” before “there exists $x \in S \setminus \{a\}$ ”) and the deleted-point set $S \setminus \{a\}$ present. 1 mark if the quantifier order is reversed or $S \setminus \{a\}$ is written as S .
- **(3 marks)** Key steps of the proof: 1 mark for correctly negating the definition of limit point for A (obtaining ε_1); 1 mark for the same for B (obtaining ε_2); 1 mark for taking $\varepsilon = \min\{\varepsilon_1, \varepsilon_2\}$ and correctly concluding that no point of $(A \cup B) \setminus \{x\}$ lies within ε of x .
- **(1 mark)** The three writing expectations: complete sentences, statements joined by connectives, and inference shown with “therefore”/“hence”/“thus”/“so” or $\therefore$. Deduct this mark for a proof written as a bare symbolic chain.

Typical deductions: using the Sequential Criterion for Limit Points despite the prohibition (0 marks for the proof); forgetting to exclude x itself when forming $A \setminus \{x\}$ or $B \setminus \{x\}$; taking $\varepsilon = \varepsilon_1 + \varepsilon_2$ or $\max\{\varepsilon_1, \varepsilon_2\}$ instead of the minimum; asserting the contrapositive conclusion without proving it.

Question 4

(6 Marks)

Write down the definition of a (sequentially) compact set, and **use the definition to show** that if K_1 and K_2 are compact sets, then $K_1 \cup K_2$ is compact. *Do not use the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem.*

Solution

Definition ((Sequentially) Compact Set). $K \subseteq \mathbb{R}$ is (sequentially) compact if every sequence in K has a convergent subsequence with limit in K .

Method 1: Splitting the index set

Proof. Let $\{a_n\}$ be a sequence in $K_1 \cup K_2$ and $S_i := \{n \in \mathbb{N} : a_n \in K_i\}$ for $i = 1, 2$. $S_1 \cup S_2 = \mathbb{N}$ ($a_n \in K_1 \cup K_2$ for all n) is infinite $\therefore S_1$ or S_2 is infinite (a union of two finite sets is finite); WLOG S_1 is infinite. List $S_1 = \{n_1 < n_2 < n_3 < \dots\} \therefore \{a_{n_k}\}$ is a subsequence of $\{a_n\}$ lying in K_1 . K_1 compact $\therefore a = \lim a_{n_{k_j}}$ for some subsequence $\{a_{n_{k_j}}\}$, with $a \in K_1$. $j \mapsto n_{k_j}$ is strictly increasing (composition of strictly increasing maps) $\therefore \{a_{n_{k_j}}\}$ is a subsequence of $\{a_n\}$. $\therefore \{a_n\}$ has a convergent subsequence with limit $a \in K_1 \subseteq K_1 \cup K_2 \therefore K_1 \cup K_2$ is compact. $\square$

Marking scheme (6 marks)

QRS suggested marking scheme.

- **(2 marks)** Correct definition of a (sequentially) compact set: every sequence in K has a convergent subsequence with limit in K . 1 mark if “in K ” is dropped from either the subsequence’s convergence or its limit.
- **(3 marks)** Key steps: 1 mark for correctly splitting the index set $\mathbb{N}$ into two pieces and arguing at least one is infinite; 1 mark for constructing the subsequence $\{a_{n_k}\}$ lying entirely in one K_i with strictly increasing indices; 1 mark for correctly applying the compactness of that K_i to extract the final convergent subsequence and noting it is still a subsequence of $\{a_n\}$.
- **(1 mark)** The three writing expectations: complete sentences, connectives, and explicit inference (“therefore”/“hence”/“thus”/“so” or $\therefore$).

Typical deductions: citing the Sequential Compactness of Closed and Bounded Sets or the Bolzano-Weierstrass Theorem (0 marks for the proof, since both are prohibited); asserting without justification that “infinitely many terms lie in K_1 or K_2 ”; forgetting to check that the extracted sub-subsequence is a genuine subsequence of the original $\{a_n\}$; omitting “without loss of generality” when only one case is argued.